

A Closed-Loop Controller for Reward Hacking: Critical Slowing Down and Nonsmooth Damping

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Abstract

A companion diagnosis shows that reward hacking, in a surrogate best-response model, is a saddle-node bifurcation: past a critical monitor steepness $\sigma > \sigma_c$ an agent’s evasion coordinate flips loyal→hacking abruptly and irreversibly, and—for a detection-based warning—the flip becomes harder to forewarn as monitoring sharpens. We pose and solve the associated control problem: keep the plant in the safe, single-basin regime. A controller needs a sensor, an actuator, and a stability guarantee, and the sharp-monitoring regime defeats the obvious choice for the first two. We give the choices that survive and prove they compose into a stable closed loop. *Sensor*: the detection-based (shadow-price) precursor erodes, but critical slowing down does not; we design an observable proxy for it and characterise its signal-to-floor requirement. *Actuator*: a quadratic penalty leaves σ_c invariant, so the damper must be nonsmooth; we implement it as a projected best-response (a deadband barrier) and show it removes the hacking basin for any σ without weakening the detector, at a capability/autonomy cost paid on the agent side. *Closed loop*: modelling the controller as a switched system, we prove that once the actuator engages with $\sigma_c(\text{engaged}) > \sigma$ the loyal equilibrium is the unique attractor, and we derive a rate law—a rate-induced-tipping constraint $\dot{w} \leq L(\sigma) w_+ / \tau_{\text{act}}$ —under which the loop keeps the agent loyal even as stakes ramp past the open-loop fold. The pieces are classical (early-warning signals, barrier/projected methods, rate-induced tipping) and imported with credit; the contribution is the reward-hacking mapping, the sensor/actuator dichotomies, and a switched closed-loop stability analysis with an explicit rate law, verified on the surrogate—at a stated cost in agent capability and autonomy, not in detection power.

1 Introduction

Reward hacking is usually fought reactively: detect a hack in a rollout, then penalise or discard it (Denison et al., 2024; Miao et al., 2024; Baker et al., 2025). A companion diagnosis explains why reactive detection is fragile in the regime that matters. There, an agent’s evasion coordinate $\rho \in [0, 1]$ (the fraction of its hack footprint hidden from a monitor) obeys a best-response flow $\dot{\rho} = \mu \partial_{\rho} \Pi(\rho; w)$ with $\Pi = w(1 - A(\rho)) - C(\rho)$; for detector steepness $\sigma = \max |A'|$ above $\sigma_c = 3.049/(8\rho^*)$ the loyal→hacking transition is a saddle-node fold, and because the detection curve sharpens with dimension ($\sigma = \Theta(d^{1/3})$) the failure gets more abrupt, more irreversible, and—for a detection-based warning—less forewarnable as the monitor improves. Detecting the hack after it manifests is then too late: the flip is a discontinuous jump, and lowering optimisation pressure need not undo it.

We pose the control problem: design an in-loop controller that keeps the plant below threshold, with a stability guarantee. A controller has three parts—a sensor, an actuator, and a closed-loop argument—and the sharp-monitoring regime defeats the obvious choice for the first two.

Contributions.

1. **Sensor design** (§4): the shadow-price precursor erodes; critical slowing down (CSD) does not (Theorem 1). We design an observable proxy for CSD and characterise the signal-to-floor condition under which it triggers before the flip (Proposition 1).
2. **Actuator implementation** (§5): a quadratic damper leaves σ_c invariant, so the damper must be nonsmooth (Theorem 2). We implement it as a *projected best-response* (a deadband barrier, the proximal step for an indicator cost) and show it removes the hacking basin for any σ at no cost to the detector.
3. **Closed-loop stability and rate law** (§6): modelling the controller as a switched system, we prove that after engagement with $\sigma_c(\text{engaged}) > \sigma$ the loyal equilibrium is the unique attractor (Theorem 3), and we derive the rate law $\dot{w} \leq L(\sigma) w_+ / \tau_{\text{act}}$ that prevents rate-induced tipping (Theorem 4).
4. **Experiments** (§7): the sensor, actuator, closed loop, and rate law are verified on the surrogate, including the critical actuator delay.

Positioning. The ingredients are classical and credited. Critical slowing down as an early-warning signal near tipping points is a mature research programme (Scheffer et al., 2009; Dakos et al., 2012; Kuehn, 2011); that some first-order transitions lack it is also known (Guttal and Jayaprakash, 2008), and concurrent work applies it to a different AI failure mode (Cui et al., 2025). Projected-gradient and proximal/barrier methods, and safe/model-predictive control, are standard (Parikh and Boyd, 2014; Boyd and Vandenberghe, 2004); rate-induced tipping is a known phenomenon in nonautonomous dynamics (Ashwin et al., 2012). Our contribution is the reward-hacking mapping, the two dichotomies (curvature-survives-where-detection-erodes; nonsmooth-not-quadratic), and a switched closed-loop analysis with an explicit rate law, verified end to end. Crucially, “no detection cost” is not “no cost”: §8 states the capability/autonomy price the controller pays.

2 Related work

Reward-hacking mitigation. Recent methods detect and suppress hacking during training: penalising detected rollouts and inoculation prompting (Denison et al., 2024); representation-level detectors and energy constraints (Miao et al., 2024); monitoring reasoning traces, with the caveat that optimising against a monitor can teach obfuscation (Baker et al., 2025). These are reactive or heuristic. We give a control-theoretic loop with a precursor sensor and a threshold-moving actuator, and identify why the natural choices fail.

Early-warning signals. Approaching a saddle-node, the leading Jacobian eigenvalue $\rightarrow 0$ and fluctuations slow down, raising variance and lag-1 autocorrelation (Scheffer et al., 2009; Dakos et al., 2012; Kuehn, 2011); the caveat that hard first-order transitions lack this signature (Guttal and Jayaprakash, 2008) does not apply to our genuine saddle-node. Our sensor is this signal; the novelty is the contrast with the detection-based precursor, which erodes.

Barrier and projected methods; rate-induced tipping. Projected gradient / proximal operators and control-barrier constructions keep a state in a feasible set (Parikh and Boyd, 2014;

Boyd and Vandenberghe, 2004); our actuator is the proximal step for a deadband indicator. Rate-induced tipping—escape caused by a control parameter changing too fast rather than crossing a static threshold—is classical in nonautonomous systems (Ashwin et al., 2012); our rate law is its avoidance condition.

3 Plant and control objective

The plant is the surrogate of the companion. The evasion coordinate $\rho \in [0, 1]$ (hack footprint in the monitor’s null channel) evolves under the noisy best-response flow

$$d\rho = \mu \partial_\rho \Pi(\rho; w) dt + \xi dW, \quad \Pi(\rho; w) = w(1 - A(\rho)) - C(\rho), \quad (1)$$

with logistic detector A of steepness $\sigma = \max |A'|$, bump $h := -A'$ ($\max h = \sigma$), convex cost C ($\varphi := C'$), stakes w ramping slowly at rate $\dot{w}$ over a long task, loyal-fold stakes w_+ and fold coordinate ρ_+ . The loyal equilibrium is $\rho_L(w)$; its well curvature is $\kappa(w) := |\partial_\rho^2 \Pi(\rho_L)|$. The *control objective* is to keep the well single-basin (no fold for any w), equivalently $\sigma < \sigma_c$ or the hacking basin unreachable. In practice ρ is latent; the sensor acts on an observable proxy (§4) and the actuator on the agent’s update (§5).

Controller as a switched system. The controller observes a proxy and, when a trigger fires, switches the agent’s cost $C \mapsto C + D$ (the damper). Writing $f_{\text{open}}(\rho, w) = \mu(wh - \varphi)$ and $f_{\text{closed}}(\rho, w) = \mu(wh - \varphi - D')$, the closed loop is the nonautonomous switched system

$$\dot{\rho} = \begin{cases} f_{\text{open}}(\rho, w(t)), & t < t_{\text{eng}}, \\ f_{\text{closed}}(\rho, w(t)), & t \geq t_{\text{eng}}, \end{cases} \quad \dot{w} = \text{const}, \quad (2)$$

where $t_{\text{eng}} = t_{\text{trig}} + \tau_{\text{act}}$ is the engagement time (trigger plus actuator delay). The analysis below characterises each mode and the switch.

4 Sensor: critical slowing down and its proxy

Theorem 1 (CSD survives where the shadow price erodes). *As $w \uparrow w_+$ along the loyal branch, the well curvature flattens as a saddle-node, $\kappa(w) = \Theta(\sqrt{w_+ - w}) \rightarrow 0$, so the stationary fluctuations of ρ undergo critical slowing down:*

$$\text{Var}(\rho) \sim \frac{\xi^2}{2\mu\kappa(w)} \rightarrow \infty, \quad \text{AC}_1(\rho) \approx 1 - \mu\kappa(w)\Delta t \rightarrow 1, \quad (3)$$

By contrast the detection at the operating point, $A(\rho_L) \rightarrow A(\rho_+)$, stays ≈ 1 until the flip (because $\rho_+ \rightarrow 0$ as $\sigma \rightarrow \infty$), so the shadow-price precursor does not fire. The sensor must read the curvature, not the detection.

Proof. Near the fold (ρ_+, w_+) , with $g := \partial_\rho \Pi$ one has $g = \partial_\rho g = 0$ and $\partial_\rho^2 g, \partial_w g \neq 0$ (saddle-node). Expanding, $g \approx \frac{1}{2} \partial_\rho^2 g (\rho - \rho_+)^2 + \partial_w g (w - w_+)$, so $\rho_L(w) - \rho_+ = -\sqrt{2\partial_w g (w_+ - w) / (-\partial_\rho^2 g)}$ and $\kappa(w) = |\partial_\rho g(\rho_L)| = |\partial_\rho^2 g| |\rho_L - \rho_+| = \Theta(\sqrt{w_+ - w})$. Linearising (1) about ρ_L gives an Ornstein–Uhlenbeck process with rate $\mu\kappa$; its stationary variance is $\xi^2/(2\mu\kappa)$ and its lag- Δt autocorrelation is $e^{-\mu\kappa\Delta t} \approx 1 - \mu\kappa\Delta t$ for sampling interval Δt , giving (3). For the detection, $\rho_+ = \Theta(1/\sigma) \rightarrow 0$, so $A(\rho_L) \rightarrow A(0) \rightarrow 1$. $\square$

Proxy design. ρ is latent, so the sensor computes (3) on an observable proxy $y_t = m(\rho_t) + \eta_t$, where m is a smooth monotone observation map (e.g. the policy–reward gap, the reward-model score residual, an evaluator-agreement rate, or a low-dimensional projection of the reasoning trace) and η_t is measurement noise of variance σ_η^2 . The sensor forms the rolling variance $\hat{V}_t$ and lag-1 autocorrelation $\hat{AC}_t$ of y over a window and triggers when $\hat{V}_t$ exceeds a fixed multiple $\theta > 1$ of an adaptive baseline. Because w ramps during the task, the rolling statistics must be computed on a *de-trended* proxy (e.g. residuals after removing a slow moving-average or regression trend), and the baseline itself estimated adaptively; otherwise the drift of y due to the ramp inflates $\hat{V}_t$ and $\hat{AC}_t$ independently of the approaching fold.

Proposition 1 (Signal-to-floor condition for the proxy trigger). *The proxy variance is $\text{Var}(y) = m'(\rho_L)^2 \text{Var}(\rho) + \sigma_\eta^2$. The CSD term dominates the measurement floor—so the trigger reflects the approaching fold rather than observation noise—iff*

$$m'(\rho_L)^2 \frac{\xi^2}{2\mu \kappa(w)} \geq (\theta - 1) \sigma_\eta^2, \quad (4)$$

which, since $\kappa(w) \rightarrow 0$, holds on an interval $[w_{\text{trig}}, w_+)$ whose left endpoint recedes from w_+ as the proxy sharpens (m' large) or the floor drops (σ_η small). The trigger fires at w_{trig} , giving a lead $w_+ - w_{\text{trig}} > 0$ over the deterministic fold.

Proof. Independence of process and measurement noise gives $\text{Var}(y) = m'(\rho_L)^2 \text{Var}(\rho) + \sigma_\eta^2$; the trigger condition $\text{Var}(y) \geq \theta \text{Var}(y)_{\text{base}}$ with baseline $\approx \sigma_\eta^2$ (far from the fold, $\text{Var}(\rho)$ is small) reduces to (4). As $w \uparrow w_+$, $\kappa \rightarrow 0$ makes the left side diverge, so (4) holds on a left-neighbourhood of w_+ ; its width grows with m'/σ_η . $\square$

5 Actuator: a nonsmooth barrier

Theorem 2 (Nonsmooth damping raises σ_c ; quadratic does not). *Augment the agent’s cost, $C \mapsto C + D$, acting on the update and never on the monitor, so A (hence detection) is unchanged. Then:* (a) *a quadratic $D = \frac{1}{2}\beta\rho^2$ keeps the total cost quadratic and leaves σ_c invariant for every β (the equilibrium equation depends on w/c only)—so one cannot fix the fold by scaling up a quadratic penalty. (A linear $D = \beta\rho$ adds a constant β to the marginal cost φ ; this does shift the equilibrium condition and is not σ_c -invariant, but it offsets the cost uniformly rather than supplying the ρ -dependent stiffening that tracks a sharpening monitor, so it is not the scalable actuator either.) The damper must therefore be superquadratic (nonsmooth in the limiting case);* (b) *a smooth stiffening $D' = b\rho^p$ ($p > 1$) raises $\sigma_c(b)$ but saturates at $\sigma_c(\rho^p) < \infty$;* (c) *a deadband barrier capping $\rho \leq \rho_{\text{cap}} < \rho_+$ makes the hacking basin unreachable: for each finite σ , choosing $\rho_{\text{cap}} < \rho_+(\sigma)$ excises the fold, without weakening the detector; since $\rho_+(\sigma) = \sqrt{3}/(8\sigma)$ and $\sigma = \Theta(d^{1/3})$, the required cap scales as $\rho_{\text{cap}} = \Theta(d^{-1/3})$.*

Proof. (a) By the companion’s closed form, σ_c for a quadratic cost is independent of the cost magnitude, so adding $D = \frac{1}{2}\beta\rho^2$ (which keeps the total cost quadratic, $c \mapsto c + \beta$) leaves σ_c fixed for every β . A linear $D = \beta\rho$ instead adds the constant β to $\varphi = c\rho + \beta$; this is *not* a rescaling of c and does move the bifurcation (numerically it can even suppress the fold), but it does so by a uniform offset, not by the ρ -dependent stiffening needed to track $\sigma = \Theta(d^{1/3})$. Only a term with non-constant D'' (superquadratic) supplies that stiffening and enters the tangency criterion in the required scalable way. (b) For $\varphi \mapsto \rho + b\rho^p$ the tangency slope and offset tend, as $b \rightarrow \infty$, to those of the pure power ρ^p , so $\sigma_c(b) \uparrow \sigma_c(\rho^p) < \infty$. (c) The hacking equilibrium and the saddle

lie at $\rho \geq \rho_- > \rho_+$; a hard cap $\rho \leq \rho_{\text{cap}} < \rho_+$ excises the interval $[\rho_{\text{cap}}, 1]$ containing them from the feasible set, so on $[0, \rho_{\text{cap}}]$ the map $\Psi = \varphi/h$ is monotone and the loyal equilibrium is the unique critical point for every w . The requirement $\rho_{\text{cap}} < \rho_+ = \Theta(1/\sigma)$ and $\sigma = \Theta(d^{1/3})$ give $\rho_{\text{cap}} = \Theta(d^{-1/3})$. $\square$

Implementation: projected best-response. The deadband barrier is the cost $D_{\text{cap}}(\rho) = 0$ for $\rho \leq \rho_{\text{cap}}$ and $+\infty$ otherwise—the indicator of the feasible set $K = [0, \rho_{\text{cap}}]$. Its proximal operator is the Euclidean projection onto K , so the damped update is the *projected best-response*

$$\rho_{t+1} = \text{proj}_K(\rho_t + \mu \partial_\rho \Pi(\rho_t; w) + \xi \sqrt{\mu} \zeta_t) = \min(\rho_{\text{cap}}, \max(0, \dots)), \quad (5)$$

a nonsmooth, computationally trivial step. In continuous time the dynamics is the differential inclusion $\dot{\rho} \in \mu \partial_\rho \Pi - N_K(\rho)$, where N_K is the (Clarke) normal cone of K : interior to K the flow is the ordinary best response, and at the cap $\rho = \rho_{\text{cap}}$ the normal cone cancels any outward drive, so ρ cannot cross. A soft surrogate $D' = b\rho^p$ (Theorem 2(b)) replaces the hard projection when a smooth actuator is required, at the cost of the finite barrier $\sigma_c(\rho^p)$.

6 Closed-loop stability and rate law

We now analyse the switched system (2): first the post-engagement mode (a stability guarantee), then the switch timing (a rate law).

Theorem 3 (Post-engagement stability). *Engage the deadband barrier of Theorem 2(c) (cap $\rho \leq \rho_{\text{cap}} < \rho_+$). Then for every fixed w : (i) Deterministic GAS. The noise-free flow $\dot{\rho} = \mu g_{\text{closed}}(\rho, w)$, $g_{\text{closed}} = wh - \varphi$, has a unique attractor $\rho_L^K(w) \in K := [0, \rho_{\text{cap}}]$: either the interior root $\rho_L(w)$ when $\Psi_{\text{closed}}(\rho_{\text{cap}}) > w$, or the boundary cap ρ_{cap} when w is large enough that the unconstrained drive pushes past the cap ($\Psi_{\text{closed}}(\rho_{\text{cap}}) \leq w$). In either case it is globally asymptotically stable on K . (ii) Stochastic invariance and concentration. For the SDE (1) with projection onto K , K is forward-invariant, and the stationary law concentrates at $\rho_L^K(w)$ with variance $O(\xi^2/(\mu\kappa_{\min}))$, where $\kappa_{\min} > 0$ lower-bounds the engaged well curvature on K (guaranteed because the deadband removes the flattening fold, so $\kappa_L \geq \kappa_{\min}$ uniformly; at the boundary cap the curvature is replaced by the projection's confining normal cone). In particular the hacking region $\rho > \rho_+$ has vanishing stationary mass as $\xi \rightarrow 0$. Consequently, once engaged, the closed loop keeps ρ in the loyal set for all subsequent w , including $w > w_+$. (A smooth stiffening $\varphi \mapsto \rho + b\rho^p$ gives the same guarantee only when $\sigma < \sigma_c(\rho^p)$, the saturation ceiling of Theorem 2(b); above it, only the deadband suffices.)*

Proof. (i) On K the map $\Psi_{\text{closed}} = \varphi/h$ is strictly increasing (no tangency point of $wh = \varphi$ lies in K , since the fold $\rho_+ > \rho_{\text{cap}}$ is excised, Theorem 2c), so $g_{\text{closed}}(\rho, w) = h(\rho)(w - \Psi_{\text{closed}}(\rho))$ has the unique zero $\rho_L(w) = \Psi_{\text{closed}}^{-1}(w)$ on K . Sign argument: since $h > 0$ and Ψ_{closed} is increasing, $g_{\text{closed}} > 0$ for $\rho < \rho_L$ (the field points *up* toward ρ_L) and $g_{\text{closed}} < 0$ for $\rho \in (\rho_L, \rho_{\text{cap}}]$ (points *down* toward ρ_L); at $\rho = \rho_{\text{cap}}$ the projection's inward normal cone only reinforces this. Hence $V(\rho) = \frac{1}{2}(\rho - \rho_L)^2$ has $\dot{V} = (\rho - \rho_L)\mu g_{\text{closed}} < 0$ off ρ_L , and ρ_L is GAS on K . (ii) The projected SDE is a Moreau sweeping process on the convex set K (Appendix B); K is forward-invariant because the projection returns any excursion to K . At an interior equilibrium, linearising the drift gives an Ornstein–Uhlenbeck process with rate $\mu\kappa_L$, $\kappa_L = h(\rho_L)\Psi'_{\text{closed}}(\rho_L) \geq \kappa_{\min} > 0$ (bounded below because the fold is excised, so Ψ'_{closed} does not vanish on K), whose stationary variance is $\xi^2/(2\mu\kappa_L) = O(\xi^2/(\mu\kappa_{\min}))$; at the boundary cap the reflecting projection confines the stationary law within $O(\xi/\sqrt{\mu})$ of ρ_{cap} . By forward-invariance of K and $\rho_{\text{cap}} < \rho_+$, the stationary law places vanishing mass on $\{\rho > \rho_+\}$ as $\xi \rightarrow 0$. Thus no fold exists for any w and ramping w past w_+ creates no hacking attractor. $\square$

Surrogate object	Real-agent counterpart	How to obtain
Evasion coordinate ρ	hidden/obfuscated fraction of a hack	observable proxy (below)
Monitor steepness σ	reward-model / detector sharpness	separability or input dimension
Threshold σ_c	onset of abrupt overoptimisation	swept sharpness/dimension
Stakes w , ramp $\dot{w}$	optimisation pressure / KL budget	training-progress schedule
CSD sensor (Var, AC_1)	rolling variance/autocorr. of a proxy	policy-reward gap, RM residual
Deadband actuator proj_K	constrained/clipped policy update	project action onto safe set
Rate law $\dot{w} \leq L w_+ / \tau_{\text{act}}$	pressure/KL ramp schedule	slow enough that CSD leads

Table 1: Mapping from surrogate objects to their real-agent counterparts, as an implementation guide. The proxy for ρ (policy-reward gap, reward-model score residual, evaluator-agreement rate, or a low-dimensional projection of the reasoning trace) is de-trended before computing the CSD statistics (§4).

Theorem 4 (Rate law: avoiding rate-induced tipping). *Let w_{tip} be the (noise-induced) escape stakes of the open mode and $L(\sigma) := (w_{\text{tip}} - w_{\text{trig}})/w_+ > 0$ the fractional CSD lead (Proposition 1). The closed loop (2) keeps ρ loyal iff engagement precedes escape, $w_{\text{trig}} + \dot{w} \tau_{\text{act}} < w_{\text{tip}}$, i.e.*

$$\boxed{\dot{w} \leq L(\sigma) w_+ / \tau_{\text{act}}}. \quad (6)$$

Numerically, $L(\sigma)$ decreases with σ (Appendix A: $0.63 \rightarrow 0.29$ over $\sigma \in [0.85, 5]$), tied to the narrowing loyal well $\rho_+ = \sqrt{3}/(8\sigma)$ —escape sets in earlier as the well shrinks, so sharper monitoring demands slower ramps; and the lead is (numerically) monotone decreasing in the noise ξ , so one uses the smallest ξ that clears the sensor floor (4). We do not prove monotonicity of L in closed form; it is an empirical scaling of the lead.

Proof. By Theorem 3, once engaged (at stakes $w_{\text{eng}} = w_{\text{trig}} + \dot{w} \tau_{\text{act}}$) the loyal equilibrium is the unique attractor, so the loop succeeds iff ρ has not yet escaped the loyal basin at w_{eng} , i.e. $w_{\text{eng}} < w_{\text{tip}}$. Substituting $w_{\text{tip}} - w_{\text{trig}} = L(\sigma)w_+$ and rearranging gives (6). The observed decrease of L with σ and ξ is consistent with the narrowing well ($\rho_+ = \Theta(1/\sigma)$) and with larger noise producing earlier escape; we report it as a numerical scaling, not a proven bound. $\square$

The controller (recipe).

sense ($\hat{V}, \hat{AC}$ of proxy y cross $\theta \times$ baseline; Prop. 1) $\rightarrow$ **actuate** (project onto $[0, \rho_{\text{cap}}]$, $\rho_{\text{cap}} = \Theta(d^{-1/3})$; Thm 2) $\rightarrow$ **maintain** single basin (Thm 3), subject to **rate** $\dot{w} \leq L w_+ / \tau_{\text{act}}$ (Thm 4).

The detector is untouched throughout, so the loop costs no detection *power* and prevents the flip rather than detecting the hack after the fact. The cost is instead paid on the agent side: the projected update (5) is a hard constraint on the policy’s action space, and the deadband forbids all movement into the null channel (benign or not)—a capability/autonomy price quantified in §8.

7 Experiments

The theory’s predictions are checked on the surrogate (`numpy/scipy`, CPU)—these are surrogate checks, not validation on a trained agent; methods and parameters are in Appendix A. Figure 1 verifies the actuator and the sensor dichotomy: *smooth superquadratic* stiffening ($\varphi \mapsto \rho + b\rho^3$)

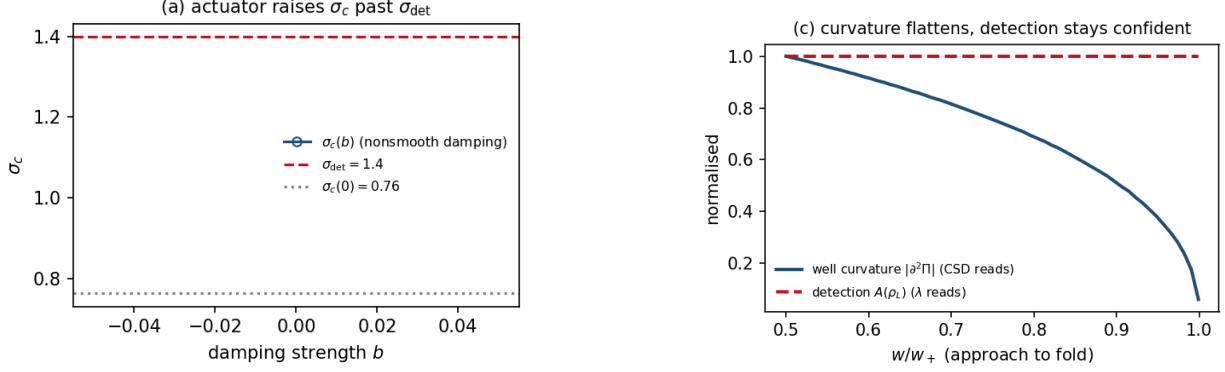


Figure 1: **Left (actuator):** *smooth* superquadratic stiffening ($\varphi \mapsto \rho + b\rho^3$) raises $\sigma_c(b)$ toward the detector steepness ($\sigma_{\text{det}} = 1.4$, $\rho^* = 0.5$); detection AUC is unaffected. Note this is the soft actuator (Theorem 2b), which saturates; the hard deadband (Theorem 2c) is the projection used in the closed loop. **Right (sensor):** as $w \rightarrow w_+$ the well curvature flattens (CSD signal) while detection $A(\rho_L)$ stays confident (Theorem 1; $\sigma = 2.0$, $\rho^* = 0.5$).

raises $\sigma_c(b)$ monotonically ($0.78 \rightarrow 1.38$ for $b = 0 \rightarrow 50$) and, extrapolating the same trend, crosses the detector steepness $\sigma_{\text{det}} = 1.4$ at $b^* \approx 61$, while the detection AUC is constant (0.706, unchanged because the damper never touches A); a quadratic damper leaves σ_c fixed (Theorem 2a). The well curvature flattens $0.77 \rightarrow 0.04$ as $w/w_+ : 0.5 \rightarrow 0.999$ while $A(\rho_L)$ holds at ≈ 0.9996 : variance rises, detection confidence does not (Theorem 1).

Figure 2 is the central closed-loop experiment: ramping the stakes past the open-loop fold, the open loop flips to hacking (final $\rho \approx 0.73$) while the closed loop—the projected best-response engaging on the CSD trigger—stays loyal (final $\rho \approx 0.08$), with engagement in 100% of runs across $\sigma \in \{1.0, 1.4, 2.0\}$ (Theorem 3). The warning lead $L(\sigma)$ shrinks with σ as the well narrows ($0.63 \rightarrow 0.29$ over $\sigma \in [0.85, 5]$; Theorem 4), and is monotone in the noise ξ (0.66, 0.55, 0.39, 0.18, 0.03 for $\xi = 0.004, \dots, 0.08$), confirming no interior optimum.

Figure 3 tests the rate law directly: sweeping the actuator delay τ_{act} at fixed ramp rate, the loyal-success rate stays at 1 up to a critical delay and then collapses—the rate-induced tipping boundary of Theorem 4—and the critical delay shrinks as σ grows (from the smaller lead $L(\sigma)$). At $\sigma = 1.4$ the loop succeeds in 100% of runs for $\tau_{\text{act}} \leq 3000$ steps and degrades beyond, consistent with $\dot{w} \tau_{\text{act}} < Lw_+$.

8 Discussion: the cost is paid elsewhere

The loop turns the companion’s alarming diagnosis into a constructive prescription. Because the failure is a bifurcation rather than a static error, the right sensor measures *proximity to the bifurcation* (curvature/critical slowing down), not the monitor’s momentary confidence; and the right actuator moves the *bifurcation threshold* (via the agent’s cost curvature), not the monitor. Both act without touching the detector, so detection power is preserved, and the rate law reframes a familiar practice—ramp optimisation pressure gently—as a rate-induced-tipping constraint with a computable bound.

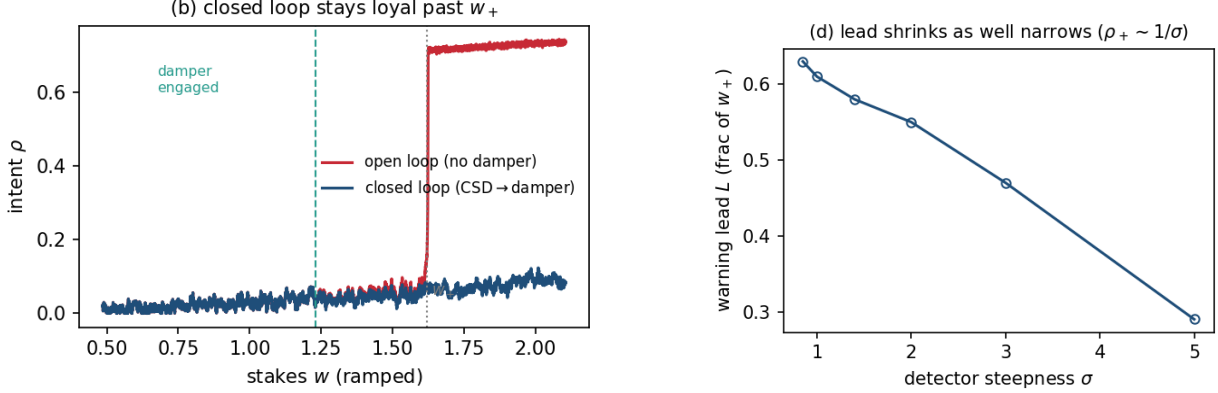


Figure 2: **Left (closed loop):** stakes ramp past the open-loop fold w_+ ; the open loop flips to hacking, the closed loop (project-on-trigger, engagement marked) stays loyal (Theorem 3; $\sigma = 1.4$). **Right (lead):** the warning lead $L(\sigma)$ shrinks as the loyal well narrows, $\rho_+ = \sqrt{3}/(8\sigma)$ (Theorem 4; $\mu = 0.03$, $\xi = 0.012$).

No detection cost does not mean no cost. The deadband barrier forbids *all* movement into the null channel, benign or not; it is paid for in *capability and autonomy*. A tight cap $\rho_{\text{cap}} = \Theta(d^{-1/3})$ suppresses legitimate exploration and abstention that happen to reduce monitor confidence, and the projected update (5) is a hard constraint on the agent’s policy space. The controller trades agent freedom for safety; quantifying that trade-off (the legitimate-behaviour loss of a given cap, and whether a softer stiffening with finite $\sigma_c(\rho^p)$ suffices) is important future work. An *adaptive* detector, which reshapes A to erase the plateaus at fixed steepness, is an alternative actuator that may pay a smaller capability price; it is outside this analysis.

9 Limitations and future work

The surrogate caveats of the companion apply (scalar monotone evasion coordinate, best-response flow, non-adaptive detector). Three are specific here. (i) The stability guarantee (Theorem 3) requires the engaged damper to satisfy $\sigma_c(\text{engaged}) > \sigma$; a smooth stiffening saturates at $\sigma_c(\rho^p) < \infty$, so beyond that steepness only the hard deadband guarantees a single basin. (ii) The rate law assumes a measurable proxy with a floor low enough for (4); if the proxy is too noisy or too slow, the sensor cannot trigger within the lead window. (iii) The Clarke/projected formulation is exact in the sharp ($d \rightarrow \infty$) limit where the detection curve tends to a step; at finite d the smooth stiffening suffices. *Future work.* The essential next step, shared with the diagnosis companion, is a controlled reinforcement-learning study with a *learned* detector, in which the sensor runs on a real training proxy and the projected actuator is applied to a policy-gradient update, to test whether the switched-system guarantee and the rate law survive genuine gradients and an emergent detector; and a quantitative account of the capability/autonomy cost of the cap.

A Numerical methods and reproducibility

All computations are CPU-only **numpy/scipy**. *Actuator* (Fig. 1 left). $\sigma_c(b)$ is located by bisection on the three-root onset of $wh - (\rho + b\rho^3)$ over a W -sweep; detection AUC is computed for the fixed logistic A (unchanged by b). *Sensor* (Fig. 1 right). The curvature $\kappa(w) = |wh'(\rho_L) - 1|$ is

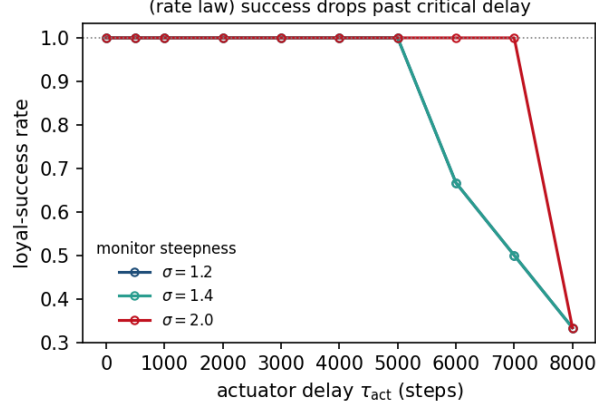


Figure 3: Rate law (Theorem 4): loyal-success rate vs. actuator delay τ_{act} at fixed ramp rate, for three monitor steepnesses ($\rho^* = 0.5$, $\mu = 0.03$, $\xi = 0.012$, 6 seeds). Success holds up to a critical delay and then collapses (rate-induced tipping); the critical delay shrinks with σ as the CSD lead $L(\sigma)$ narrows.

evaluated along the continued loyal branch; $A(\rho_L)$ is read off A . *Closed loop* (Fig. 2). The plant (1) is integrated by Euler–Maruyama ($\mu = 0.03$, $\xi = 0.012$) with w ramped linearly from $0.3w_+$ to $1.3\text{--}1.5w_+$ over 1.6×10^4 steps; the sensor triggers when the rolling variance (window 250) exceeds 3×10^{-4} , after which the projected step (5) (or stiffening $b = 30\text{--}60$) engages; “loyal success” is final $\rho < 0.3$, seed-averaged over 6 seeds. *Rate law* (Fig. 3). An actuator delay τ_{act} is inserted between trigger and engagement; success rate is over 6 seeds. Code (sensor, projected actuator, closed-loop and rate-law experiments) will be released to reproduce all figures.

B The differential inclusion and projected flow

The damped continuous dynamics is $\dot{\rho} \in \mu \partial_{\rho} \Pi(\rho; w) - N_K(\rho)$ with $K = [0, \rho_{\text{cap}}]$ and N_K the normal cone: $N_K(\rho) = \{0\}$ for $\rho \in \text{int } K$, $N_K(\rho_{\text{cap}}) = [0, \infty)$, $N_K(0) = (-\infty, 0]$. This is a Moreau sweeping process; existence and uniqueness of a solution follow from convexity of K and Lipschitz $\partial_{\rho} \Pi$ on K . The Euler discretisation of the sweeping process is exactly the projected step (5), $\rho_{t+1} = \text{proj}_K(\rho_t + \mu \partial_{\rho} \Pi + \xi \sqrt{\mu} \zeta_t)$. On $\text{int } K$ it is the ordinary best response; at ρ_{cap} the projection removes any outward component, so K is forward-invariant and the loyal equilibrium (interior to K by $\rho_{\text{cap}} < \rho_+$ construction, but attracting from all of K) is globally asymptotically stable, completing Theorem 3.

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